Exam 1 Review

Intro to Audio Tech 1
Jocelyn Kavanagh
Wednesday, September 17

Georgia Tech

Exam Topics

- Sinusoids
  - Frequency, amplitude, and phase
  - Perceptual properties
  - How phase shifts impact the audio output (constructive/destructive interference)
  - $a\cos(\omega t + \phi)$
- Amplitude and Loudness
  - Human perception of loudness
    - Equal-Loudness contours
  - Know the RMS equation
    - $RMS = \sqrt{mean(amplitude^2)}$
    - Rule of thumb for sinusoids: $RMS = a_peak/(\sqrt{2})$
  - Know the decibel power equation
    - $dB = 20 \cdot log(a/a_0)$
- Frequency
  - Human perception of frequency
  - Harmonics
  - The relationship between frequency and period
    - T=1/f

Georgia Tech

Exam Topics

- Digital Audio
  - Sampling Theorem
    - Nyquist Frequency -> max freq/2
  - Aliasing
    - $f_o = f - N * f_s$
  - Sampling Rate and Bit Depth
- Classical Waveforms and Spectra
  - How to build them
  - Magnitude Spectrum
- Max
  - Be comfortable with the following objects: cycle~, message, bang, math operators, metro, trigger, number, line~
  - Know how to control amplitude, frequency, phase
  - Recognize errors in a patch

Georgia Tech

Exam Format

- Canvas
- Full class time
- 25 points
  - 11 multiple choice, matching, true/false
  - 2 open-ended questions (one is math, the other is theory and explanation)
  - 2 open-ended Max patch questions
- You need to memorize your equations
- You may use a calculator and scratch paper

Georgia Tech

Example Questions

Classical Waveforms

- All harmonics with strength 1/n
  - Saw
- Odd harmonics with strength 1/n
  - Square
- Odd harmonics with strength 1/n^2
  - Triangle
- Single frequency strength of 1
  - Sinusoid

Calculate Period and Phase

- A sinusoid has an initial phase of 0 and angular frequency (radians/sample) of $\omega = \frac{\pi}{10}$. What is the period in samples? What is the phase at sample n=10?

$$N = \frac{2\pi}{\omega} = \frac{2\pi * 10}{\pi} = 20\ samples$$

$$\theta[n] = \omega n + \phi = \frac{\pi}{10} * 10 + 0 = \pi$$

Calculate Period and Phase

- A sinusoid has an initial phase of 0 and angular frequency (radians/sample) of $\omega = \frac{\pi}{10}$. What is the period in samples? What is the phase at sample n=10?

$$N = \frac{2\pi}{\omega} = \frac{2\pi * 10}{\pi} = 20\ samples$$

$$\theta[n] = \omega n + \phi = \frac{\pi}{10} * 10 + 0 = \pi$$

Calculate Decibels

- If 0 dB corresponds to an amplitude of 1, how many dB corresponds to amplitudes of 1.5, 2, 3, and 5?

$$0 = 20\log(1)$$

$$20\log(1.5) = 3.52$$

$$20\log(2) = 6.02$$

$$20\log(3) = 9.54$$

$$20\log(5) = 13.979$$

Georgia Tech

Relationship between frequency and SR

- What is the angular frequency of middle C at 44100 samples per second?

Middle C is 261.6 Hz

Angular frequency (radians/sample) = $\omega = 2\pi \frac{f}{f_s}$

$$\omega = 2\pi \frac{261.6}{44100} = 0.0373 \text{ rad/sample}$$

Harmonics and Sampling Rate

- How many harmonics of a tone at A 440 can be represented digitally at a sampling rate of 44100 Hertz?

Nyquist frequency is maximum frequency = SR/2 = 44100/2 = 22050 Hz

Harmonics are integer multiples, so divide the bandwidth allowance by fundamental

22050/440 = 50.11 so 50 harmonics

Sampling Rate and Aliasing

- What frequency would you hear if you synthesized a sinusoid at 88000 Hertz at a sampling rate of 44100?

88,000Hz is above the Nyquist frequency, 22050Hz, so there will be aliasing

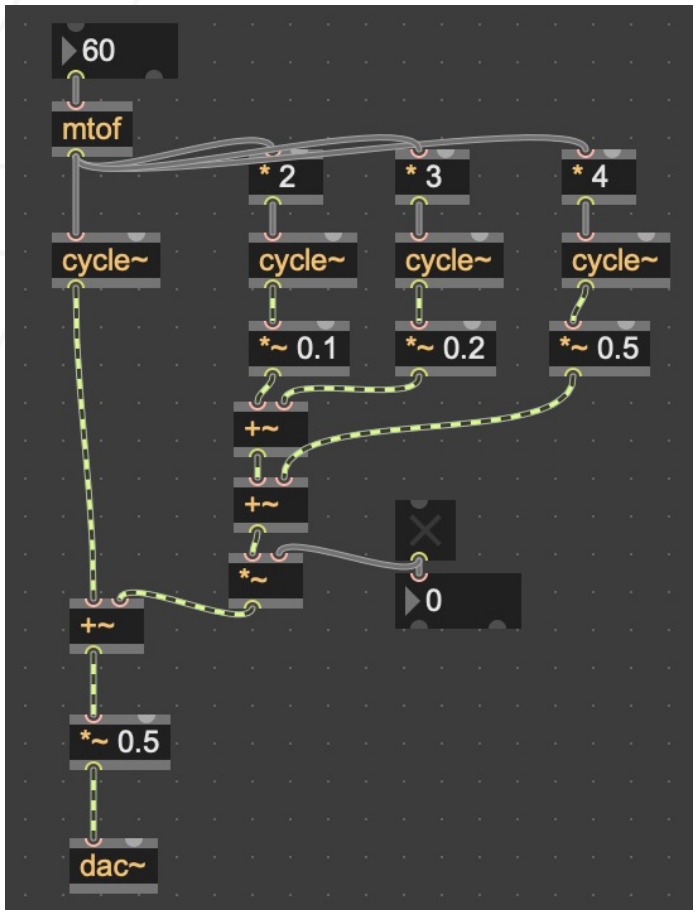
$$f_o = f - N * f_s$$

$$N = round\left(\frac{f}{f_s}\right) = \frac{88000}{44100} = 1.995$$

$$f_o = f - N * f_s = 88000 \ - 2 * 44100 = -200$$

Georgia Tech

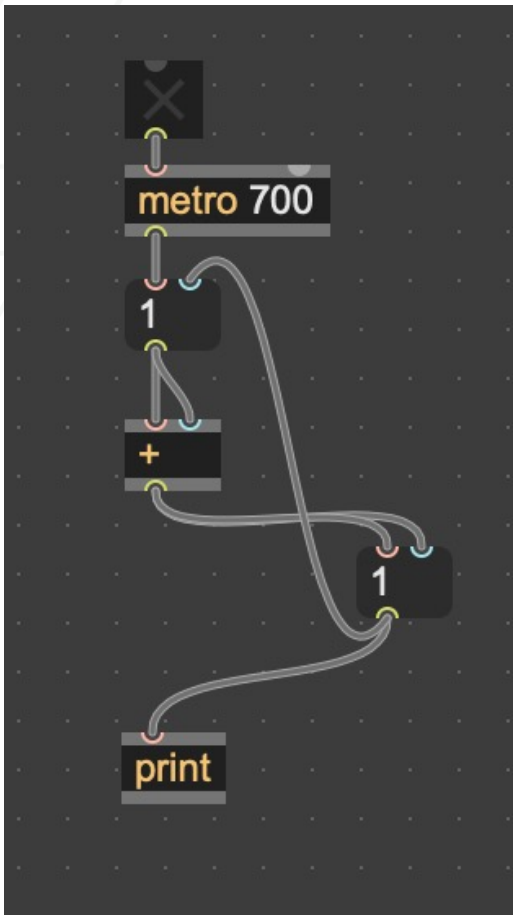
Describe the Max Patch



- An additive synth with 4 harmonics
- Takes in midi notes and flips to frequency
- Each harmonic's amplitude is attenuated
- When the toggle is pressed, the harmonics are present otherwise they are silent

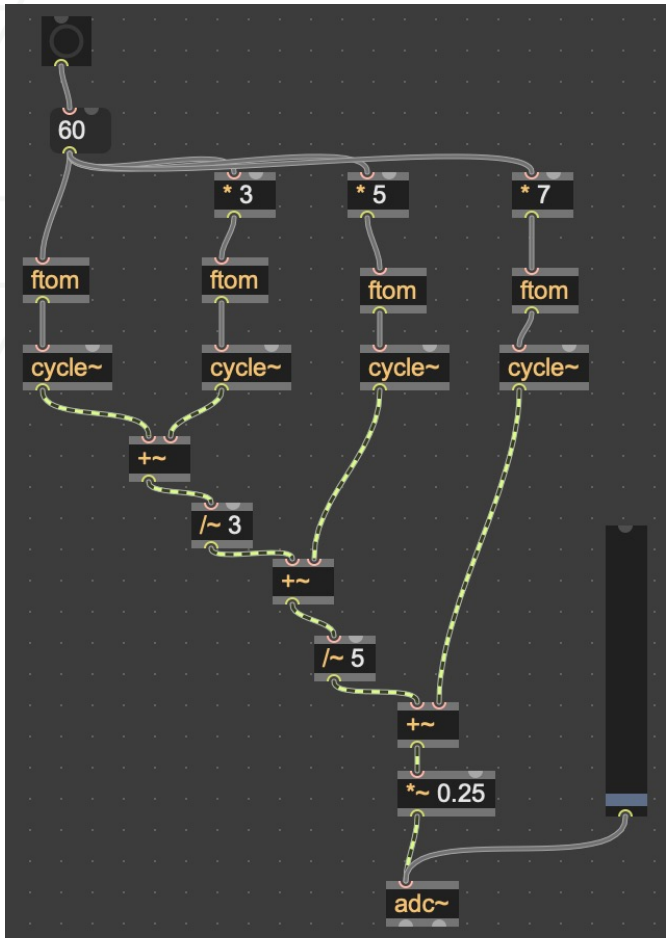
Georgia Tech

Describe the Max Patch



- When the toggle is pressed, every 700ms the first message will be added to itself and printed
- So, the first few values will be 2, 4, 8, 16

Fix the Max Patch



- I'm trying to create a triangle wave with midi input. I'm testing middle C first.
- I also want overall amplitude control

- What's wrong with this patch?

Georgia Tech